



CSE 445

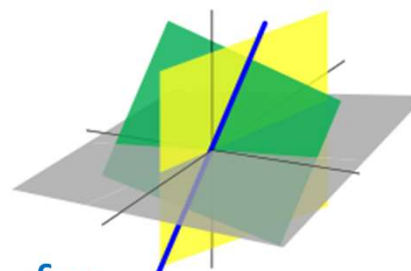
Lecture 2-3

Linear Algebra



What is Linear Algebra

- Linear algebra is the branch of mathematics concerning linear equations such as
 - $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$
 - In vector notation we say $a^T x = b$
 - Called a linear transformation of x
- Linear algebra is fundamental to geometry, for defining objects such as lines, planes, rotations
 - Linear equations $a^T x = b$ defines a plane in $(x_1, x_2, \dots, x_n)$ space
 - Straight lines define common solutions to equations



Why Linear Algebra?

- Linear Algebra is used throughout engineering
 - Because it is based on continuous math rather than discrete math
 - Traditional CS curriculum embracing it increasingly
- Essential for understanding ML algorithms
 - We convert input vectors $(x_1, x_2, \dots, x_n)$ into outputs by a series of linear transformation
- In this course we discuss
 - Concepts of linear algebra needed for ML
 - Omit other aspects of linear algebra



Topics Covered

- Scalars, Vectors, Matrices, Tensors
- Matrix & Vector multiplication
- Linear dependence & Span
- Norms
- Special matrices/vectors
- Eigen decomposition
- Singular Value Decomposition (SVD)
- Moore–Penrose pseudoinverse
- Trace & Determinant
- Example: Principal Components Analysis (PCA)

Key concepts: Scalar

- Single number
 - Just a number
 - In contrast to other objects in linear algebra, which are usually arrays of numbers
- Represented in lower-case italic x
 - They can be real-valued or be integers
 - Let $x \in \mathcal{R}$ be the slope of the line
 - Defining a real valued scalar
 - Let $\eta \in \mathcal{N}$ be the number of units
 - Defining a natural number scalar

Key concepts: Vectors

- An array of numbers arranged in order
- Each number identified by an index
- Written in lower-case bold such as $\mathbf{x}$
 - Its elements are in italics lower case subscripted
- If each element is in $\mathbb{R}$ then $\mathbf{x}$ is in $\mathbb{R}^n$
- We can think of vectors as points in space
 - Each element gives coordinate along an axis

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Key concepts: Matrices

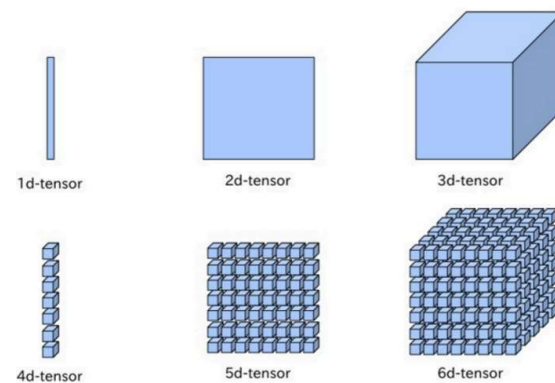
- 2-D array of numbers
 - So each element identified by two indices
- Denoted by bold typeface **A**
 - Elements indicated by name in italic but not bold
 - $A_{1,1}$ is the top left entry and $A_{m,n}$ is the bottom right entry
 - We can identify numbers in vertical column j by using ":"

$$A = \begin{bmatrix} A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,1} & A_{3,2} & A_{3,3} \end{bmatrix}$$

- $A_{i,:}$ is i^{th} row of A , $A_{:,j}$ is the j^{th} column of A
- If A has shape of height m and width n with real-values then $A \in \mathbb{R}^{m \times n}$

Key concepts: Tensors

- Sometimes need an array with more than two axes
 - An RGB color image has three axes
- A tensor is an array of numbers arranged on a regular grid with variable number of axes
- Denote a tensor with this bold typeface: **A**
- Element (i, j, k) of tensor denoted by $A_{i,j,k}$



Vectors as special case of matrix

- Vectors are matrices with a single column
- Often written in-line using transpose

$$\mathbf{x} = [x_1, \dots, x_n]^T$$

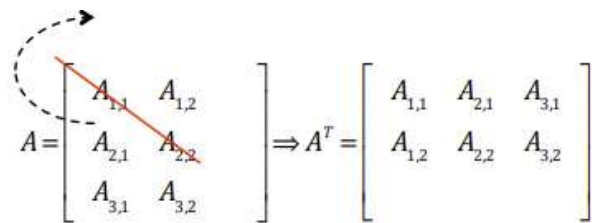
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \Rightarrow \mathbf{x}^T = [x_1, x_2, \dots, x_n]$$

- A scalar is a matrix with one element
 - $a = a^T$

Transpose of a Matrix

- An important operation on matrices
- The transpose of a matrix **A** is denoted as **A^T**
- Defined as
 - $(A^T)_{i,j} = A_{j,i}$
 - The mirror image across a diagonal line
 - Called the main diagonal running down to the right starting from upper left corner

$$A = \begin{bmatrix} A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,1} & A_{3,2} & A_{3,3} \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \\ A_{1,3} & A_{2,3} & A_{3,3} \end{bmatrix}$$



$$A = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

Matrix Addition/Subtraction

- We can add matrices to each other if they have the same shape, by adding corresponding elements
 - If A and B have the same shape (height m, width n)
 - $C = A + B \rightarrow C_{i,j} = A_{i,j} + B_{i,j}$
- A scalar can be added to a matrix or multiplied by a scalar
 - $D = aB + c \rightarrow D_{i,j} = aB_{i,j} + c$
- Less conventional notation used in ML
 - Vector added to matrix
 - $C = A + b \rightarrow C_{i,j} = A_{i,j} + b_j$
 - Called broadcasting since vector b added to each row of A

Multiplying Matrices

- For product $C=AB$ to be defined, A must have same number of columns as the number of rows of B
- IF A is of shape $m \times n$ and B is of shape $n \times p$ then matrix product C is of shape $m \times p$

$$C = AB \Rightarrow C_{i,j} = \sum_k A_{i,k} B_{k,j}$$

- Note that the standard product of two matrices is not just the product of two individual elements
 - Such a product does exist and is called the element wise product or Hadamard product

$$\mathbf{A} \odot \mathbf{B}$$

Multiplying Vectors

- Dot product between two vectors x and y of same dimensionality is the matrix product $x^T y$
- We can think of matrix product $C=AB$ as computing C_{ij} - the dot product of row i of A and column j of B

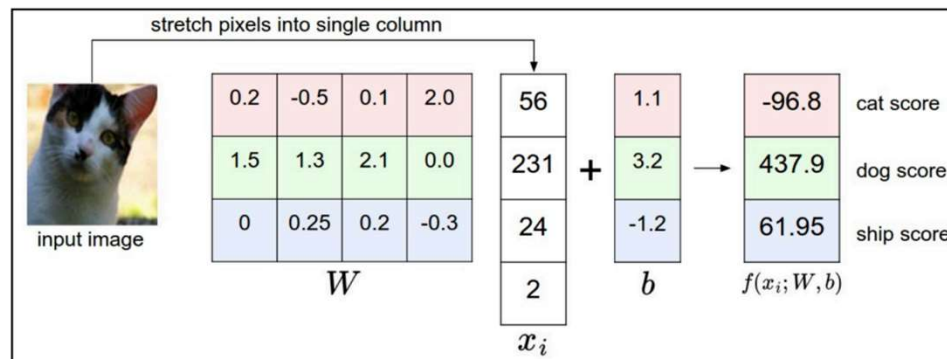
Matrix Product Properties

- Distributivity over addition: $A(B+C) = AB + AC$
- Associativity: $A(BC) = (AB)C$
- Not commutative: $AB = BA$ is not always true
- Dot product between vectors is commutative: $x^T y = y^T x$
- Transpose of a matrix product has a simple form: $(AB)^T = B^T A^T$

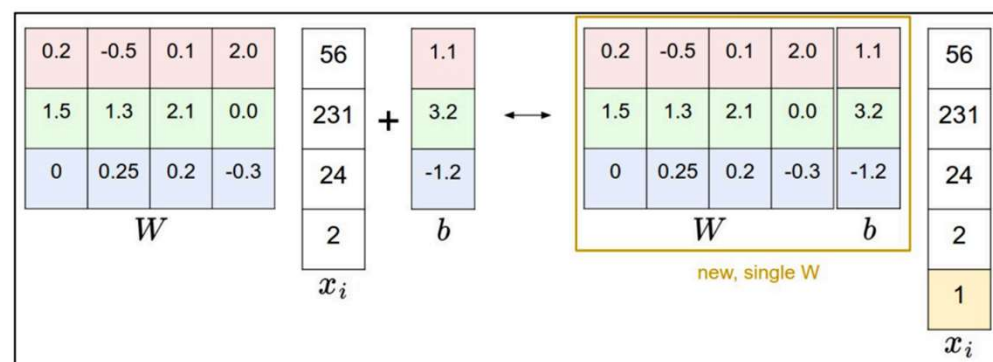
Example of tensors in ML

Vector x is converted into vector y by multiplying x by a matrix W

A linear classifier $y = Wx^T + b$



A linear classifier with bias eliminated $y = Wx^T$



Linear Transformations

- $Ax=b$

- where $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$

- More explicitly

$$A_{11}x_1 + A_{12}x_2 + \dots + A_{1n}x_n = b_1$$

$$A_{21}x_1 + A_{22}x_2 + \dots + A_{2n}x_n = b_2$$

$$A_{n1}x_1 + A_{n2}x_2 + \dots + A_{nn}x_n = b_n$$

n equations in
 n unknowns

$$A = \begin{bmatrix} A_{1,1} & \dots & A_{1,n} \\ \vdots & \vdots & \vdots \\ A_{n,1} & \dots & A_{nn} \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

$n \times n \qquad n \times 1 \qquad n \times 1$

Can view A as a *linear transformation*
of vector x to vector b

- Sometimes we wish to solve for the unknowns $x = \{x_1, \dots, x_n\}$ when A and b provide constraints

Span of a set of vectors

- Span of a set of vectors: set of points obtained by a *linear combination* of those vectors
 - A linear combination of vectors $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$ with coefficients c_i is $\sum_i c_i \mathbf{v}^{(i)}$
 - System of equations is $\mathbf{Ax}=\mathbf{b}$
 - A column of $\mathbf{A}$, i.e., $\mathbf{A}_{:,i}$ specifies travel in direction i
 - How much we need to travel is given by x_i
 - This is a linear combination of vectors $\mathbf{Ax} = \sum_i x_i \mathbf{A}_{:,i}$
 - Thus determining whether $\mathbf{Ax}=\mathbf{b}$ has a solution is equivalent to determining whether $\mathbf{b}$ is in the span of columns of $\mathbf{A}$
 - This span is referred to as *column space* or *range* of $\mathbf{A}$

Span of a set of vectors

- If you have one nonzero vector in $\mathbb{R}^2$
 - Its span is a line through the origin in the direction of that vector.
- If you have two linearly independent vectors in $\mathbb{R}^2$
 - Their span is the entire plane $\mathbb{R}^2$
- If you have two linearly independent vectors in $\mathbb{R}^3$
 - Their span is a plane through the origin in $\mathbb{R}^3$
- If you have three linearly independent vectors in $\mathbb{R}^3$
 - Their span is the whole 3D space.
- Connection to Machine Learning
 - The span tells us what subspace of the input space can be represented by a set of vectors.
 - In regression, for example, the span of the feature vectors determines what outputs can be represented exactly.
 - In PCA, the top k eigenvectors span the low-dimensional subspace used for dimensionality reduction.
- **In short: The span is the “space you can reach” by combining vectors. It’s like the coverage area formed by scaling and adding your vectors together.**

Linearly dependent/independent

- Linear Independence

- A set of vectors (or matrix columns) is linearly independent if no vector can be written as a linear combination of the others.
- Example: In $\mathbb{R}^2$, vectors $(1,0)$ and $(0,1)$ are independent.

- Linear Dependence

- A set of vectors (or matrix columns) is linearly dependent if at least one vector can be expressed as a linear combination of others.
- Example: In $\mathbb{R}^2$, vectors $(1,2)$ and $(2,4)$ are dependent because $(2,4) = 2 \times (1,2)$.

- Matrix Perspective

- Columns (or rows) of a matrix are independent if the rank = number of columns (or rows).
- If rank < number of columns, then some columns are dependent.

Norm

- Definition

- A matrix norm is a function that assigns a non-negative value (size/length) to a matrix, similar to vector norms.
- A norm is a function that measures the length or magnitude of a vector.

- Properties of a Norm

For any matrix A :

- $\|A\| \geq 0$ and $\|A\| = 0 \iff A = 0$
- $\|\alpha A\| = |\alpha| \|A\|$ (scaling)
- $\|A + B\| \leq \|A\| + \|B\|$ (triangle inequality)
- **Submultiplicative:** $\|AB\| \leq \|A\| \|B\|$

For any vector x, y and scalar α :

- **Non-negativity:** $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$
- **Homogeneity:** $\|\alpha x\| = |\alpha| \|x\|$
- **Triangle Inequality:** $\|x + y\| \leq \|x\| + \|y\|$

L^p Norm

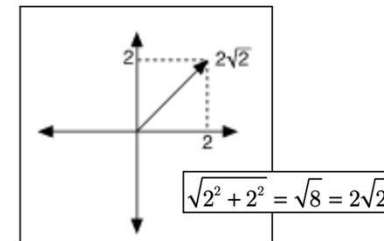
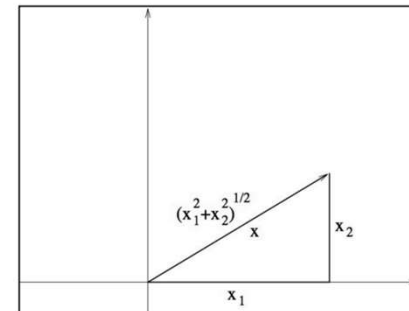
- Definition:

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{\frac{1}{p}}$$

- L^2 Norm

- Called Euclidean norm

- Simply the Euclidean distance between the origin and the point x
- written simply as $\|x\|$
- Squared Euclidean norm is same as $x^T x$



- L^1 Norm

- Useful when 0 and non-zero have to be distinguished

- Note that L^2 increases slowly near origin, e.g., $0.1^2=0.01$)

- L^∞ Norm $\|x\|_\infty = \max_i |x_i|$

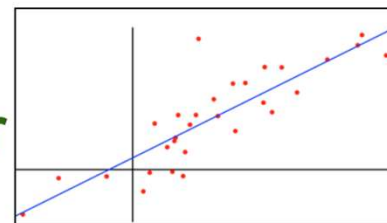
- Called max norm

Use of norm in Regression

- Linear Regression

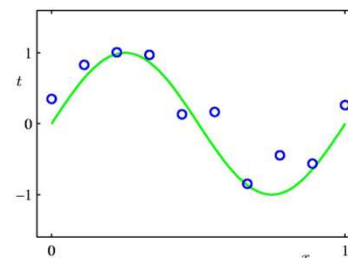
$\mathbf{x}$: a vector, $\mathbf{w}$: weight vector

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + \dots + w_d x_d = \mathbf{w}^T \mathbf{x}$$



With nonlinear basis functions ϕ_j

$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\mathbf{x})$$



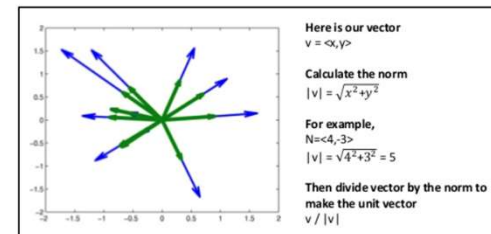
- Loss Function

$$\tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(\mathbf{x}_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2} \|\mathbf{w}^2\|$$

Second term is a weighted norm
called a regularizer (to prevent overfitting)

L^P Norm and Distance

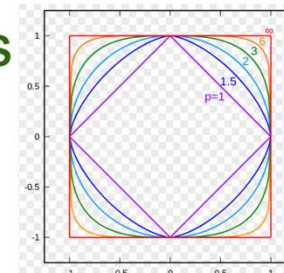
- Norm is the length of a vector



- We can use it to draw a unit circle from origin

– Different P values yield different shapes

- Euclidean norm yields a circle



- Distance between two vectors (v, w)

– $\text{dist}(v, w) = ||v - w||$

$$= \sqrt{(v_1 - w_1)^2 + \dots + (v_n - w_n)^2}$$

Distance to origin would just be sqrt of sum of squares

Size of a Matrix

- **Definition**

The Frobenius norm of a matrix $A \in \mathbb{R}^{m \times n}$ is:

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$$

- **Intuition**

- Treats the **matrix as a vector** of all its entries.
- Equivalent to the **Euclidean norm** of the flattened matrix.

Properties

- Always non-negative: $\|A\|_F \geq 0$
- Unitary invariant: $\|UAV\|_F = \|A\|_F$ for orthogonal/unitary U, V
- Related to singular values:

$$\|A\|_F = \sqrt{\sum_i \sigma_i^2}$$

Example

For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \|A\|_F = \sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{30}$$

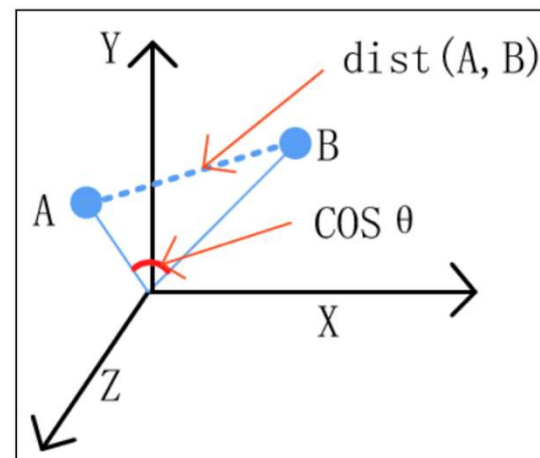
Angle between Vectors

- Dot product of two vectors can be written in terms of their L^2 norms and angle θ between them

$$\mathbf{x}^T \mathbf{y} \Rightarrow \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos \theta$$

- Cosine between two vectors is a measure of their similarity

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$



Special kind of Matrix: Diagonal

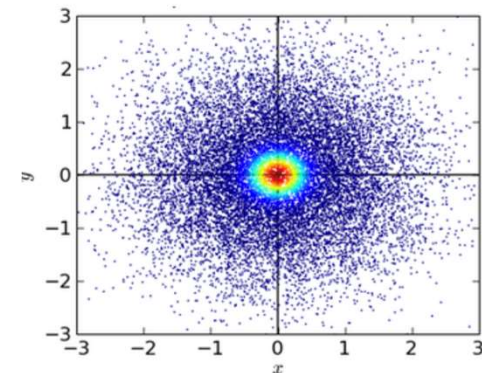
- Diagonal Matrix has mostly zeros, with non-zero entries only in diagonal
 - E.g., identity matrix, where all diagonal entries are 1

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
 - E.g., covariance matrix with independent features

$$\text{Cov}(X, Y) = \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\begin{aligned} \text{Covariance} &= \frac{\sum (x_i - x_{\text{avg}})(y_i - y_{\text{avg}})}{n-1} \\ \text{Covariance} &= \frac{-64.57}{8} \\ \text{Covariance} &= -8.07 \end{aligned}$$

$$\begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_K^2 \end{bmatrix}$$



If $\text{Cov}(X, Y) = 0$ then $E(XY) = E(X)E(Y)$

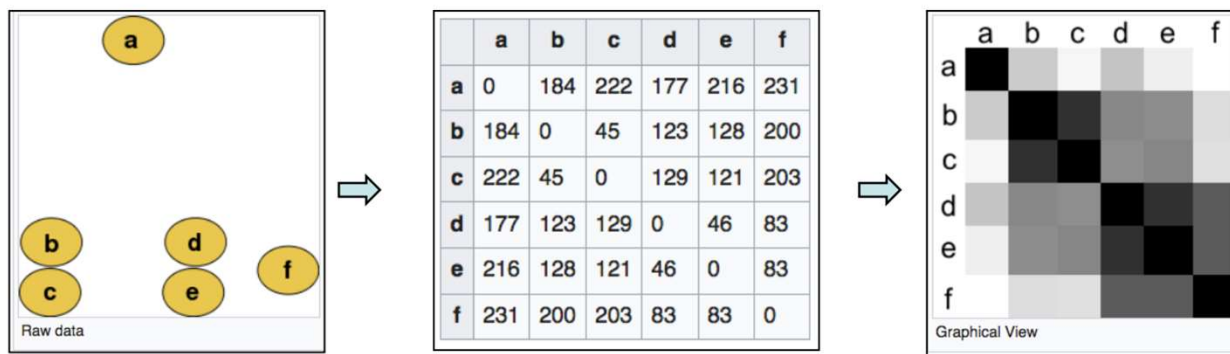
$$N(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Efficiency of Diagonal Matrix

- $\text{diag}(\mathbf{v})$ denotes a square diagonal matrix with diagonal elements given by entries of vector $\mathbf{v}$
 - Multiplying vector $\mathbf{x}$ by a diagonal matrix is efficient
 - To compute $\text{diag}(\mathbf{v})\mathbf{x}$ we only need to scale each x_i by v_i
- $$\text{diag}(\mathbf{v})\mathbf{x} = \mathbf{v} \odot \mathbf{x}$$
- Inverting a square diagonal matrix is efficient
 - Inverse exists iff every diagonal entry is nonzero, in which case $\text{diag}(\mathbf{v})^{-1} = \text{diag}([1/v_1, \dots, 1/v_n]^T)$

Symmetric Matrix

- A symmetric matrix equals its transpose: $A = A^T$
 - E.g., a distance matrix is symmetric with $A_{ij} = A_{ji}$



- E.g., covariance matrices are symmetric

$$\Sigma = \begin{pmatrix} 1 & .5 & .15 & .15 & 0 & 0 \\ .5 & 1 & .15 & .15 & 0 & 0 \\ .15 & .15 & 1 & .25 & 0 & 0 \\ .15 & .15 & .25 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & .10 \\ 0 & 0 & 0 & 0 & .10 & 1 \end{pmatrix}$$

Special Kinds of Vectors

- Unit Vector

- A vector with unit norm

$$\|x\|_2 = 1$$

- Orthogonal Vectors

- A vector x and a vector y are orthogonal to each other if $x^T y = 0$

- If vectors have nonzero norm, vectors at 90 degrees to each other

- Orthonormal Vectors

- Vectors are orthogonal & have unit norm

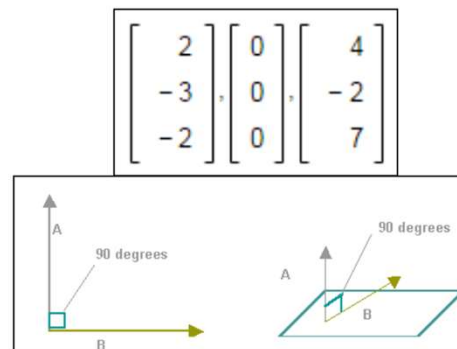
- Orthogonal Matrix

- A square matrix whose rows are mutually

orthonormal: $A^T A = A A^T = I$

- $A^{-1} = A^T$

Orthogonal matrices are of interest because their inverse is very cheap to compute



Basis

- Vector basis: Coordinates in finite dimensions.
- Function basis: Building blocks for infinite-dimensional function spaces.

- A **basis function** is a **fundamental building block** used to construct other functions.
- A set of basis functions must be:
 - **Linearly independent** (none is redundant)
 - **Spanning** (any function in the space can be expressed as their combination)

Mathematically, for a function space V , a set $\{\phi_1(x), \phi_2(x), \dots, \phi_n(x)\}$ is a **basis** if:

$$f(x) = c_1\phi_1(x) + c_2\phi_2(x) + \dots + c_n\phi_n(x)$$

for some coefficients c_i .

- **Definition**

A **basis** is a set of vectors/functions that are:

- **Linearly independent**
- Can **span** the entire space

- **Basis Functions in Vector Spaces**

- Standard basis in $\mathbb{R}^n$:

$$e_1 = (1, 0, 0, \dots), e_2 = (0, 1, 0, \dots), \dots$$

- Any vector can be expressed as a **linear combination** of basis vectors.

- **Polynomial Basis**

$$\{1, x, x^2, \dots, x^n\}$$

- Used in polynomial regression, curve fitting.
- Any polynomial function can be expressed as their linear combination.

- **Fourier Basis**

$$\{\sin(nx), \cos(nx)\}, \quad n = 1, 2, \dots$$

- Used for representing **periodic functions/signals**.
- Allows decomposition of signals into frequency components.

Determinant of a Matrix

- Determinant of a square matrix $\det(A)$ is a mapping to a scalar
- It is equal to the product of all eigenvalues of the matrix
- Measures how much multiplication by the matrix expands or contracts space

Trace of a Matrix

- Trace operator gives the sum of the elements along the diagonal

$$Tr(A) = \sum_{i,j} A_{i,i}$$

- Frobenius norm of a matrix can be represented as

$$\|A\|_F = \left(Tr(A^T A)\right)^{\frac{1}{2}}$$

Matrix Decomposition

- Matrices can be decomposed into factors to learn universal properties, just like integers:
 - Properties not discernible from their representation

1. Decomposition of integer into prime factors

- From $12 = 2 \times 2 \times 3$ we can discern that
 - 12 is not divisible by 5 or
 - any multiple of 12 is divisible by 3
 - But representations of 12 in binary or decimal are different

2. Decomposition of Matrix A as $A = V \text{diag}(\lambda) V^{-1}$

- where V is formed of eigenvectors and λ are eigenvalues, e.g,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

has eigenvalues $\lambda=1$ and $\lambda=3$ and eigenvectors V :

$$v_{\lambda=1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, v_{\lambda=3} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

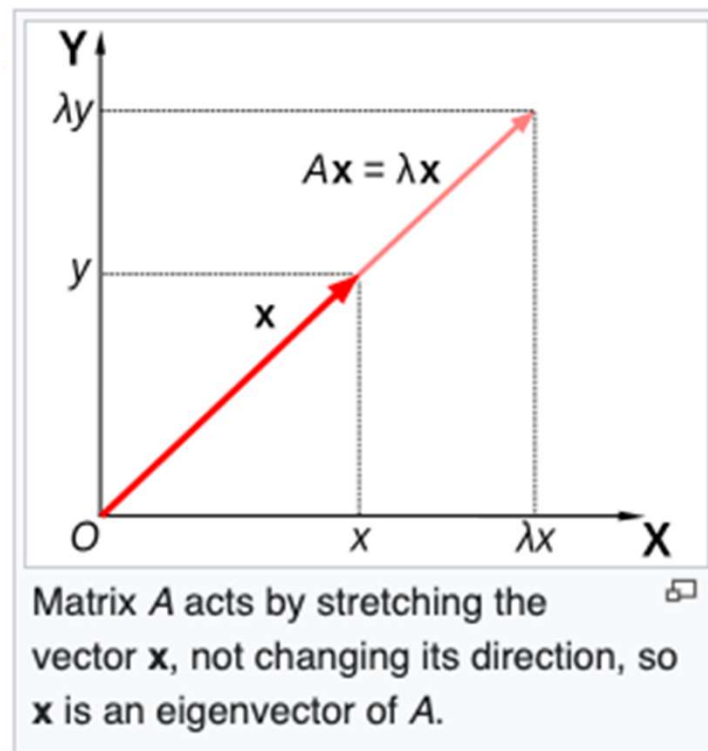
Eigenvector

- An eigenvector of a square matrix A is a non-zero vector v such that multiplication by A only changes the scale of v

$$Av = \lambda v$$

– The scalar λ is known as eigenvalue

- If v is an eigenvector of A , so is any rescaled vector sv . Moreover sv still has the same eigen value. Thus look for a unit eigenvector



Characteristic Polynomial

- Consider $Av=w$

$$A = \begin{bmatrix} A_{1,1} & L & A_{1,n} \\ M & M & M \\ A_{n,1} & L & A_{nn} \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ M \\ v_n \end{bmatrix} \quad w = \begin{bmatrix} w_1 \\ M \\ w_n \end{bmatrix}$$

- If v and w are scalar multiples, i.e., if $Av=\lambda v$
 - then v is an eigenvector of the linear transformation A and the scale factor λ is the eigenvalue corresponding to the eigen vector
- This is the *eigenvalue equation* of matrix A
 - Stated equivalently as $(A-\lambda I)v=0$
 - This has a non-zero solution if $|A-\lambda I|=0$ as
 - The polynomial of degree n can be factored as

$$|A-\lambda I| = (\lambda_1-\lambda)(\lambda_2-\lambda)\dots(\lambda_n-\lambda)$$
 - The $\lambda_1, \lambda_2, \dots, \lambda_n$ are roots of the polynomial and are eigenvalues of A

Eigenvalue/Eigenvector

- Consider the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$
- Taking determinant of $(A - \lambda I)$, the char poly is

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = 3 - 4\lambda + \lambda^2$$

- It has roots $\lambda=1$ and $\lambda=3$ which are the two eigenvalues of A
- The eigenvectors are found by solving for $\mathbf{v}$ in $A\mathbf{v} = \lambda\mathbf{v}$, which are

$$\mathbf{v}_{\lambda=1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{v}_{\lambda=3} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Eigen decomposition

- Suppose that matrix A has n linearly independent eigenvectors $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$ with eigenvalues $\{\lambda_1, \dots, \lambda_n\}$
- Concatenate eigenvectors to form matrix V
- Concatenate eigenvalues to form vector $\lambda = [\lambda_1, \dots, \lambda_n]$
- Eigendecomposition of A is given by

$$A = V \text{diag}(\lambda) V^{-1}$$

Decomposition of Symmetric Matrix

- Every real symmetric matrix A can be decomposed into real-valued eigenvectors and eigenvalues

$$A = Q\Lambda Q^T$$

where Q is an orthogonal matrix composed of eigenvectors of A : $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$

orthogonal matrix: components are orthogonal or $\mathbf{v}^{(i)T}\mathbf{v}^{(j)}=0$

Λ is a diagonal matrix of eigenvalues $\{\lambda_1, \dots, \lambda_n\}$

- We can think of A as scaling space by λ_i in direction $\mathbf{v}^{(i)}$

Why do we learn eigen decomposition?

- Matrix Simplification

- Makes powers & exponentials of matrices easier.
- Example: $A^k = V\Lambda^k V^{-1}$.

- Dimensionality Reduction

- Basis of Principal Component Analysis (PCA).
- Extracts dominant features by keeping largest eigenvalues.

- Data Analysis & ML

- Spectral clustering, topic modeling, covariance analysis.

- Engineering & Physics

- Vibration modes, stability of dynamical systems.
- Quantum mechanics (state decomposition).

- Graph Theory

- Eigenvalues/eigenvectors of adjacency or Laplacian matrices reveal community structure.

Eigen decomposition is not unique

- Eigendecomposition is $A = Q\Lambda Q^T$
 - where Q is an orthogonal matrix composed of eigenvectors of A
- Decomposition is not unique when two eigenvalues are the same
- By convention order entries of Λ in descending order:
 - Under this convention, eigendecomposition is unique if all eigenvalues are unique

Positive Definite Matrix

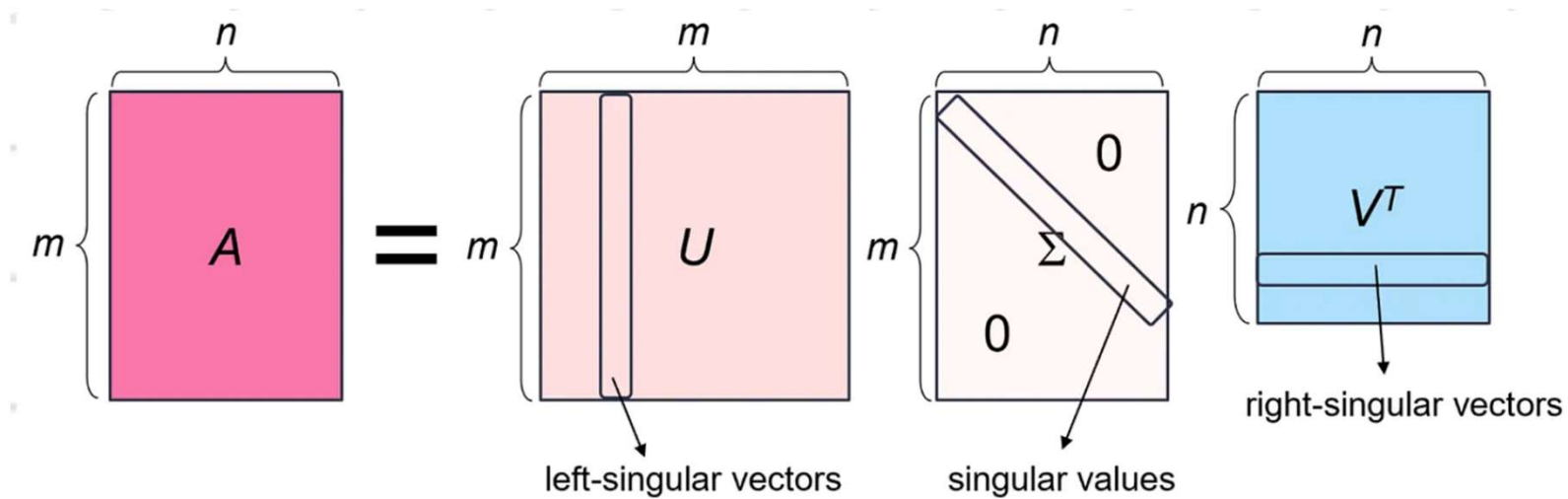
- A matrix whose eigenvalues are all positive is called *positive definite*
 - Positive or zero is called *positive semidefinite*
- If eigen values are all negative it is *negative definite*
 - Positive definite matrices guarantee that $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$

Singular Value Decomposition (SVD)

- Eigendecomposition has form: $A = V \text{diag}(\lambda) V^{-1}$
 - If A is not square, eigendecomposition is undefined
- SVD is a decomposition of the form $A = U D V^T$
- SVD is more general than eigendecomposition
 - Used with any matrix rather than symmetric ones
 - Every real matrix has a SVD
 - Same is not true of eigen decomposition

SVD

$$A = U \times \Sigma \times V^T$$



SVD Definition

- Write A as a product of 3 matrices: $A = UDV^T$
 - If A is $m \times n$, then U is $m \times m$, D is $m \times n$, V is $n \times n$
- Each of these matrices have a special structure
 - U and V are orthogonal matrices
 - D is a diagonal matrix not necessarily square
 - Elements of Diagonal of D are called *singular values* of A
 - Columns of U are called *left singular vectors*
 - Columns of V are called *right singular vectors*
- SVD interpreted in terms of *eigendecomposition*
 - Left singular vectors of A are eigenvectors of AA^T
 - Right singular vectors of A are eigenvectors of $A^T A$
 - Nonzero singular values of A are square roots of eigenvalues of $A^T A$. Same is true of AA^T